

# KARTHIK SARVAN

Rust & Embedded Systems Engineer

 [github.com/Karthik-Sarvan](https://github.com/Karthik-Sarvan)  
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## ABOUT ME

Electronics and Communication

Engineering student with a keen interest in computers, software, and embedded systems. Eager to delve into the intersection of technology and innovation, I am committed to learning and contributing to these dynamic fields. I specialize in writing safe, concurrent systems in Rust, exploring low-level kernel development, and bridging the gap between embedded hardware and AI-driven architectures.

## WORK EXPERIENCE

### Aegion Dynamics

Sept 2024 – Present  
Visakhapatnam, India

#### Embedded Intern

Gaining in-depth hands-on experience in the design, development, and optimization of embedded solutions. Focused on microcontroller-based system design with a strong emphasis on real-time hardware-software integration. Designed and analyzed electronic circuits, interfaced multiple sensors and actuators, and developed efficient low-level firmware using C and Rust ensuring reliability and performance.

### Spotmies

Sept 2024 – Present  
Visakhapatnam, India

#### MERN Stack Developer

Working as a MERN Stack Developer at REAIDY, a product-based company focused on interview preparation. My role involves developing Reaidy AI and guiding my team members. Contributing heavily to a platform designed for specialized interview practice.

### Spotmies

June 2023 – Dec 2023  
Visakhapatnam, India

#### React Intern

Completed an internship as a Web Application Developer specializing in the MERN stack. Gained production-level expertise in React, MongoDB, Git, Node.js, Express, and other related technologies while collaborating with cross-functional teams and clients.

## FEATURED PROJECTS

### Pratibim

Rust, AI, EDA Tool  
[pratibim.tech](https://pratibim.tech)

#### Core Developer

Conceptualized and developed an advanced AI-driven Electronic Design Automation (EDA) platform. The system autonomously generates circuit schematics by parsing and analyzing component datasheets, bridging the gap between AI capabilities and physical hardware design to significantly accelerate engineering workflows.

### Mimic

Robotics, C/C++  
[mimic.aegiondynamic.com](https://mimic.aegiondynamic.com)

#### Firmware Engineer | STM32 Proxy Framework

Engineered a 3-tier hardware emulation framework bridging software development and physical peripherals. Developed bare-metal C firmware on an STM32 proxy to emulate I2C, SPI, UART, and RS485 protocols, paired with a Python Bridge/CLI for real-time host control and an extensible community sensor profile ecosystem.

### Relay Test System

PCB, Automation  
Hardware Validation

#### Embedded Engineer | Diagnostic Unit

Designed a portable, automated diagnostic system for comprehensive relay validation. Engineered firmware to execute precise test sequences for contact and insulation resistance, relay bounce, non-bridging logic, and exact pickup/drop voltage thresholds.

### Defense Project

Control Systems  
Confidential

#### Firmware Engineer | Multi-Axis Actuation

Architected high-reliability firmware and developed rigorously tuned Proportional-Integral-Derivative (PID) controllers for precise multi-axis motor actuation in a secure, closed-loop defense application.

PID Balancer

Closed-Loop Logic  
Hardware Lab

Embedded Engineer | Real-Time Control

Built a real-time hardware control system utilizing custom PID algorithms to dynamically adjust platform tilt and continuously balance a freely moving object using fast-response spatial sensor feedback mapping.

EDUCATION

Chaitanya Engineering College

2023 – Present  
Pursuing B.Tech in Electronics & Communication

Gov. Polytechnic Pendurti

Completed  
Diploma in Electronics & Communication

SKILLS

**Languages:** Rust, C/C++, JavaScript, Python

**Frameworks:** MERN Stack, React, Node.js

**Systems:** Embedded Firmware, Microcontrollers, PCB Design, Linux Kernel, Control Systems (PID), Git